

Practical-1

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1. 1.1Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum is calculated using the formula: where: m is the mass of the object (in kilograms). v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Code:

```
mass = float(input()) velocity =  
float(input())  
momentum = mass*velocity print(f"{momentum:.2f}kgm/s")
```

The screenshot displays the CODETANTRA online code editor interface. At the top, the header includes the logo, navigation links (Home, Learn Anywhere), a user profile (202501050017@mitaoe.ac.in), and a Logout button. The main workspace is divided into several sections: a top status bar showing '2 out of 2 shown test case(s) passed' and '2 out of 2 hidden test case(s) passed'; a left sidebar with an Explorer view; and a central area for test cases. Test case 1 is expanded, showing 'Expected output' (5.0, 10.0, 50.00kgm/s) and 'Actual output' (5.0, 10.0, 50.00kgm/s). Test case 2 is also listed as passed. The bottom of the interface features a navigation bar with buttons for '< Prev', 'Reset', 'Submit', and 'Next >'.

1.1.2 Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer as input. Depending on the number of digits in .

Constraints:

$1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print “Invalid”.

Code:

`n=int(input()) if 0<=n`

`and n<=9:`

`print(n*n) elif`

`10<=n and n<=99:`

`print(f'{n**(1/2):.2f}') elif`

`100<=n and n<=999:`

`print(f'{n**(1/3):.2f}') else:`

`print("Invalid")`

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Explorer

Average time
0.007 s
6.57 ms

Maximum time
0.011 s
11.00 ms

4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 18 msDebug

Expected output
9
81

Actual output
9
81

Test case 25 ms

Test case 311 ms

Test case 49 ms

Debugger

< Prev

Reset

Submit

Next >

1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format YYYY-MM-DD
- The second line contains the second date in the format YYYY-MM-DD
- An integer representing the number of days between the given dates.

Code: from datetime import datetime

date1=input().strip() date2=input().strip()

d1=datetime.strptime(date1,"%Y-%m-%d")

d2=datetime.strptime(date2,"%Y-%m-%d")

difference = d2-d1 print(difference.days)

The screenshot displays the CODETANTRA web interface. At the top, the header includes the logo, navigation links (Home, Learn Anywhere), a user email (202501050017@mitaoe.ac.in), and links for Support and Logout. The main workspace is divided into several sections. On the left, an 'Explorer' sidebar shows file icons. The top status bar indicates 'Average time: 0.010 s' and 'Maximum time: 0.012 s'. A green banner at the top right of the workspace states '2 out of 2 shown test case(s) passed' and '2 out of 2 hidden test case(s) passed'. The central area shows 'Test case 1' with a '12 ms' execution time and a 'Debug' button. It compares 'Expected output' (2014-07-02, 2014-11-02, 123) with 'Actual output' (2014-07-02, 2014-11-02, 123). Below this, 'Test case 2' is shown with a '10 ms' execution time. The bottom of the interface features a dark blue bar with navigation buttons: '< Prev', 'Reset', 'Submit', and 'Next >'.

1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Code:

num = int(input()) reversed_num=

int(str(num)[::-1]) print(reversed_num)

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Explorer

Average time
0.004 s
4.00 ms

Maximum time
0.005 s
5.00 ms

2 out of 2 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 15 msDebug

Expected output
5367
7635

Actual output
5367
7635

Test case 23 ms

< PrevResetSubmitNext >

1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the

format:<number>x<multiplier>=<result> Code:

```
num = int(input()) for i in
```

```
range(1,11):
```

```
    print(f"{num} x {i} = {num * i}")
```

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Average time

0.007 s

7.00 ms

Maximum time

0.011 s

11.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

11 ms

Debug

Expected output

Actual output

8	8
8 x 1 = 8	8 x 1 = 8
8 x 2 = 16	8 x 2 = 16
8 x 3 = 24	8 x 3 = 24
8 x 4 = 32	8 x 4 = 32
8 x 5 = 40	8 x 5 = 40
8 x 6 = 48	8 x 6 = 48
8 x 7 = 56	8 x 7 = 56
8 x 8 = 64	8 x 8 = 64
8 x 9 = 72	8 x 9 = 72
8 x 10 = 80	8 x 10 = 80

Test case 2

5 ms

< Prev

Reset

Submit

Next >

1.21. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer , the number of courses.
- The second input will be integers representing the marks of the student in each of the courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- “Fail”.

Code:

```
def cal(marks, total_courses):
```

```
    if any(mark < 40 for mark in marks):
```

```

        return "Fail"

    total_marks = sum(marks)

    aggregate_percentage=(total_marks/(total_courses *100))*100    if

    aggregate_percentage >75:                                     grade="Distinction"      elif

    aggregate_percentage >=60:                                   grade= "First Division"  elif

    aggregate_percentage >=50:                                   grade= "Second Division" elif

    aggregate_percentage >=40:                                   grade= "Third Division"

    return (aggregate_percentage,grade)

num_courses= int(input()) marks=

list(map(int,input().split())) result=

cal(marks,num_courses) if result == "Fail":

    print("Fail") else:

        aggregate_percentage,grade=result    print(f'Aggregate Percentage:

{aggregate_percentage:.2f}')        print(f'Grade: {grade}')

```

The screenshot shows the CODETANTRA IDE interface. At the top, there's a navigation bar with 'Home', 'Learn Anywhere', and a user profile '202501050017@mitaoe.ac.in'. The main area displays test results for a Python program. The interface includes a left sidebar with 'Explorer' and 'Debugger' tabs. The main content area shows five test cases with their expected and actual outputs, aggregate percentages, and grades. Test cases 1-4 passed, while Test case 5 failed. The bottom bar has buttons for 'Prev', 'Reset', 'Submit', and 'Next'.

Test Case	Expected Output	Actual Output	Aggregate Percentage	Grade
Test case 1	4 55 65 78 89	4 55 65 78 89	71.75	First Division
Test case 2	5 56 78 97 86 93	5 56 78 97 86 93	82.00	Distinction
Test case 3	5 99 85 43 97 34	5 99 85 43 97 34	Fail	Fail
Test case 4	3 55 54 53	3 55 54 53	54.00	Second Division
Test case 5	5 48 45 42 48 41	5 48 45 42 48 41	43.20	Third Division

1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first terms of the Fibonacci series.

Input Format:

- A single integer representing the number of terms to generate.

Output Format:

- A single line containing the first terms of the Fibonacci sequence, separated by spaces.

```
Code: def
fibonacci(n):
    if
n == 1:
        return 0
elif n == 2:
        return 1
    else:
        return fibonacci(n - 1) + fibonacci(n - 2)
n =
int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")
```

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Explorer

Average time
0.008 s
8.25 ms

Maximum time
0.019 s
19.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 119 msDebug

Expected output
5
0 1 1 1 2 3

Actual output
5
0 1 1 1 2 3

Test case 25 ms

TerminalTest cases

< Prev

Reset

Submit

Next >

Debugger

1.23. Pattern -1

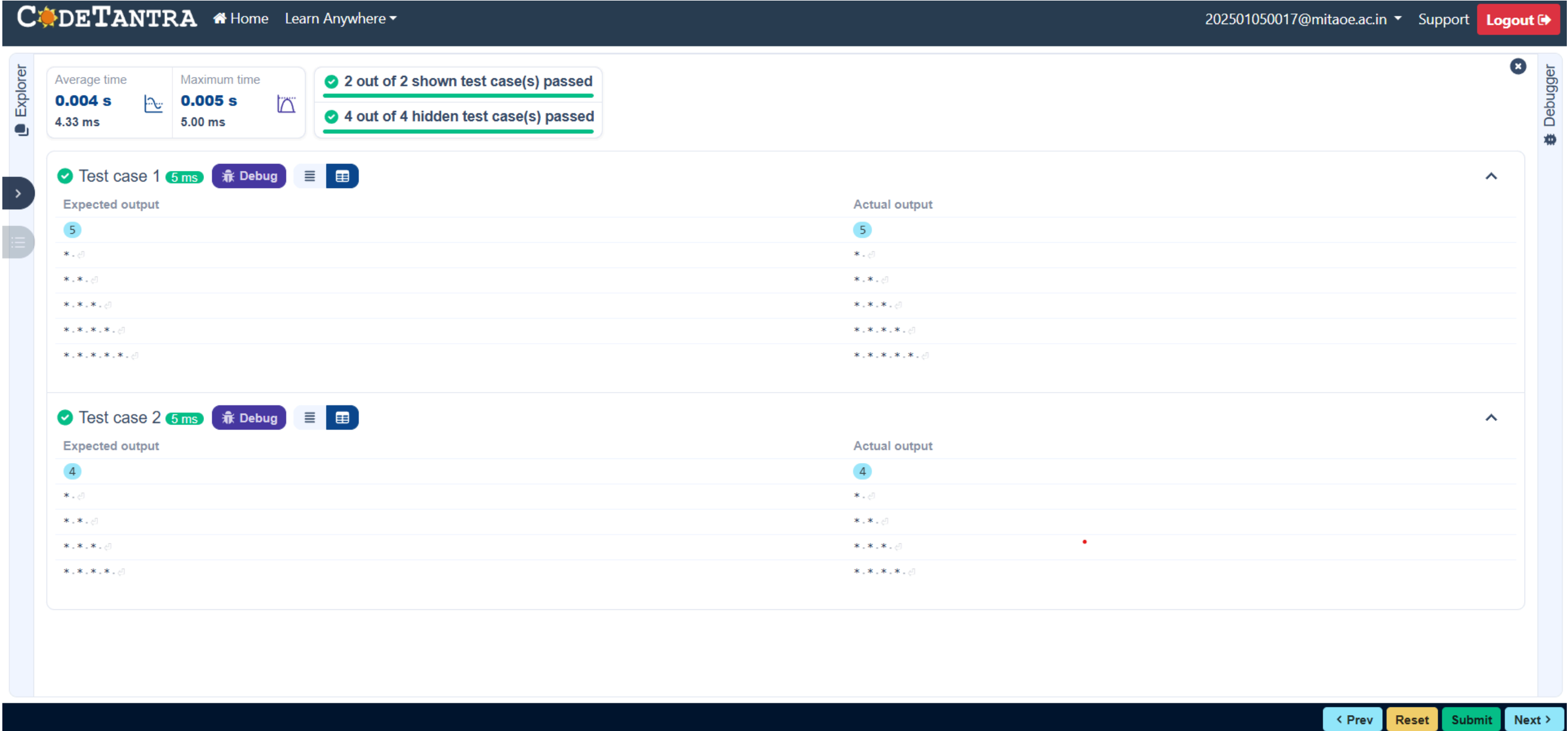
Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.
- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Code:


```
n = int(input())
for i in range (1, n+1):
    print('*' * i)
```



1.24. Pattern -2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Code:

```
n = int(input())
for i in range(1, n+1):
    for j in range(1, i+1):
        print(j,end=" ")
    print()
```


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Explorer

Average time
0.006 s
6.50 ms

Maximum time
0.008 s
8.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 16 msDebug

Expected output
5
1
1·2
1·2·3
1·2·3·4
1·2·3·4·5

Actual output
5
1
1·2
1·2·3
1·2·3·4
1·2·3·4·5

Test case 27 msDebug

Expected output
8
1
1·2
1·2·3
1·2·3·4·5
1·2·3·4·5·6
1·2·3·4·5·6·7
1·2·3·4·5·6·7·8

Actual output
8
1
1·2
1·2·3
1·2·3·4·5
1·2·3·4·5·6
1·2·3·4·5·6·7
1·2·3·4·5·6·7·8

TerminalTest cases

< Prev

Reset

Submit

Next >